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Spotify Data Analysis Report

Spotify is a digital music, podcast, and video service released in 2008, that gives you access to millions of songs and other content from creators all over the world. Music is totally free, but you can also choose to upgrade to Spotify Premium for ads free experience. This music app only gives access to music and podcasts through the application. Spotify licensing means there's no way to export music content outside of the app.
Here are a few insights into the spotify application.

- **Popularity of tracks in Spotify app.**

Most tracks fall within the 60–79 popularity range, indicating a strong concentration of moderately high-performing content. Additionally, the presence of zero-popularity tracks suggests inclusion of newly released or low-engagement songs in the catalog.
Track popularity is moderately influenced by artist popularity (r = 0.46), indicating that established artists significantly impact streaming performance.

- **Song Type distributions (*Albums/Singles/Compilation)***

-The analysis of album types indicates that most tracks (68.24%) available on Spotify belong to full-length **albums**.
-**Single** releases account for 26% of the total tracks.
-while **compilation** albums contribute only 5.91%.

This shows that most of the spotify catalogue is dominated by traditional album-based collection, very small but significantly followed by single releases.

- **Top 10 popular artist's track type**

Analysis shows that the top 10 **popular** artists are all of **single** releases rather than albums or compilations. Few of the most popular artists acquiring the spotify platforms are Taylor Swift, Drake, Justin Bieber, Rihanna, Billie Eilish , Ariana Grande.
A large **proportion** of artists on Spotify are associated with album releases, primarily because albums contain a higher number of track records compared to singles and compilations.

- **Explicit Content Coverage**

25% of Total the total catalogue consists of E labelled tracks, whereas the remaining majority catalogue comprises non-explicit content which indicates that the spotify platform is family friendly.

- **Release of tracks Yearly Analysis**

The dataset includes tracks released from 1960 to the present. However, a higher concentration of tracks originates from the 2000–2025 period, with a particularly sharp increase in releases during 2025, 2024, and 2023.

- **Yearly increase in track popularity**

Tracks released in earlier decades tend to have lower popularity scores compared to those released in recent years. From 1990 and 2000 onward, there has been a noticeable increase in average track popularity.
This upward trend may be increased due to the widespread adoption and availability of digital streaming platforms such as Spotify, which have significantly simplified music consumption and expanded audience reach.

In earlier years, accessing music was a comparatively complex process. Listeners often had to rely on physical media, limited paid platforms, or download tracks from the internet—sometimes with risks such as malware or poor-quality files.
The emergence of streaming services has eliminated many of these barriers, leading to a substantial increase in listener engagement and overall track popularity in recent years.

- **Duration of Songs**

The average duration of the songs in spotify catalogue is between 3.33-3. 49min.Tracks with the higher song duration ranging between 6.5 min to 13 min is also contained with least a number of records and preferences.

Overall, the findings reflect the growing influence of digital streaming and evolving listener behavior on Spotify.

Tools used/process followed:

- Changed the data types of respective columns for performing relevant operations.
- Checked and handled null values.
- Added and removed columns based on the analysis requirements.
- Using seaborn, matplotlib.pyplot, plotly libraries performed visual analysis like histogram, bar plot, box plot, pie chart, line plot, scattered plot.
- Used group by functions to verify and get columns-based analysis.